

# Reese Ludwick

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## Summary

AI/ML engineer who **builds and owns production MCP servers** and **video/creative-understanding AI pipelines**. Ship custom FastMCP servers and external MCP integrations giving agents dynamic tool access in production, plus a full video-understanding pipeline (autoclip.studio) that transcribes, embeds, and learns a creator's style into a scoring "DNA" module. Strong on **agentic workflows** (LangGraph multi-agent), **model-evaluation** loops, embeddings/retrieval (Voyage AI + pgvector), and FastAPI production services.

## Core Skills

- **MCP & Tooling:** Custom FastMCP servers, external MCP integrations, dynamic agent tool access, AWS-deployed
- **Video / Creative Understanding:** WhisperX transcription, Voyage AI embeddings, pgvector, ffmpeg rendering, learned style/"DNA" scoring module
- **Agentic & Eval:** LangGraph multi-agent (codegen/testing/security/supervision), LangChain, RAG, model evaluation, prompt engineering, NLP clustering, executive reporting
- **Backend & Data:** Python, FastAPI, PostgreSQL, Redis, Celery, SQL, Django REST
- **ML & Platform:** Voyage AI, Sentence Transformers, TensorFlow, Pandas, NumPy, Amazon Bedrock, Anthropic API, Docker, Git/GitHub, Linux/Ubuntu, Azure Fundamentals, React/TypeScript, Claude Code

## Experience

### Software Associate — Cognizant (Remote) · Dec 2025–Present

- Build GenAI and agentic AI systems with LangChain, LangGraph, and Python.
- **Own MCP server integrations** connecting enterprise tools and LLMs for production workflows; build RAG pipelines behind them.
- Scope, prototype, and deliver AI data tooling cross-functionally.

### GenAI Developer Contractor — Cognizant (Remote) · Jun–Dec 2025

- **Built custom FastMCP servers** and deployed external MCP integrations on AWS to give agents dynamic tool access — the core capability this role centers on.
- Ran **prompt engineering and model evaluation** to optimize LLM outputs across GenAI POCs.
- Shipped modular, production-style agentic workflows.

### Freelance AI Engineer & Full-Stack Developer — WorldCovers / APMC (Remote) · May 2026–Present

- Shipped production web properties on a 14-table PostgreSQL schema (Django REST + React/TypeScript).
- AI-augmented engineering (Claude Code) on compressed timelines with direct stakeholder coordination.

## Projects

### CreatorClip / autoclip.studio — *creative + video understanding, scoring*

*Python, FastAPI, Celery, pgvector, WhisperX, Voyage AI, Anthropic API, ffmpeg, YouTube OAuth* - AI video-clipping pipeline that **learns a creator's channel style** via a custom **"DNA" scoring module** — a direct analog to scoring *why creatives work*. - Async Celery pipeline: WhisperX transcription → Voyage AI + pgvector embeddings → ffmpeg rendering; full YouTube OAuth. - Live SaaS (Stripe billing) that turns raw footage into structured, embedded, scoreable content understanding.

### Agentic Software Development Team — *agentic workflows + evaluation*

*LangChain, LangGraph, Amazon Bedrock (Cognizant R&D)* - Multi-agent LangGraph system (codegen / testing / security / supervision) with dedicated **evaluation and security-review agents**; Bedrock + OSS ML backends.

### Customer Call Data Analysis & Insights Engine — *eval/scoring over unstructured data*

*Python, NLP, Sentence Transformers (Cognizant R&D)* - NLP clustering + embeddings grouping unstructured call data into semantic segments; agentic analysis feeding executive reporting.

### CFO Agent — *agentic financial assistant (in active development)*

*Python, FastAPI, LangGraph, PostgreSQL, Redis, Anthropic API* - FastAPI + LangGraph agent with PostgreSQL persistence, Redis caching, CSV/OFX import + encrypted vault; hybrid context architecture; multi-tenant-ready.

## Education

**BS, Computer Science — Shepherd University** · Aug 2021–May 2025 · Major GPA 3.43

## Certifications

TensorFlow Developer Specialization · Linux + Git (Boot.dev) · AZ-900 · GitHub Copilot · GenAI Context Engineering (Cognizant)